

Aerospace Academy Version

Unit 4

Measurement, Drawings & Manufacturing Documentation

Unit Overview

Students learn how engineers communicate manufacturing intent through measurement, tolerances, technical drawings, notes, bills of materials, revision control, and fabrication planning.

Essential Question

How do engineers make sure a design can be accurately manufactured, inspected, and assembled?

Name: _____ Period: _____ Date: _____

Learning Targets

1. Explain the major concepts of measurement, drawings & manufacturing documentation.
2. Apply engineering documentation, sketching, modeling, testing, or communication skills to an aerospace-focused task.
3. Use evidence and reflection to improve design work.
4. Submit a complete unit portfolio or project package.

Unit Deliverables

- Measurement practice
- Tolerance notes
- Dimensioned drawing
- Manufacturing notes
- BOM
- Revision table
- Fabrication plan
- Documentation package

Vocabulary

Term	Definition
Measurement	
Caliper	
Tolerance	
Nominal Dimension	
Actual Dimension	
Fit	
Technical Drawing	
Manufacturing Note	
Bill of Materials	
Revision	
Part Number	
Inspection	
Fabrication Plan	
Quality Control	

Unit Project: Manufacturing Documentation Package

You will apply the skills from this unit to create an aerospace-focused project package. Your work should show clear documentation, professional effort, evidence-based decisions, and reflection.

Day 1 — Measurement in Engineering

Objective: I can explain why measurement matters in manufacturing.

Bellwork: Why is “close enough” not always good enough?

Core Task: Measure sample objects with rulers/calipers and compare accuracy.

Evidence of Work / Notes:

Exit Ticket: What made measurement difficult?

Day 2 — Calipers and Precision

Objective: I can use calipers to measure dimensions.

Bellwork: What dimensions are hard to measure with a ruler?

Core Task: Measure outside, inside, depth, and thickness dimensions.

Evidence of Work / Notes:

Exit Ticket: What measurement tool was most precise?

Day 3 — Tolerance Basics

Objective: I can explain tolerance and why it matters.

Bellwork: Can every part be made perfectly? Explain.

Core Task: Analyze examples of tight and loose tolerances.

Evidence of Work / Notes:

Exit Ticket: What happens if tolerance is too loose?

Day 4 — Fits and Clearances

Objective: I can describe how parts fit together.

Bellwork: Why do holes need clearance?

Core Task: Compare sliding, press, and loose fits using simple examples.

Evidence of Work / Notes:

Exit Ticket: What kind of fit does your project need?

Day 5 — Technical Drawing Review

Objective: I can identify parts of a technical drawing.

Bellwork: What information should a drawing include?

Core Task: Identify title block, views, dimensions, notes, scale, and revision info.

Evidence of Work / Notes:

Exit Ticket: What part of a drawing is most important?

Day 6 — Dimensioning for Manufacturing

Objective: I can dimension a part so it can be made.

Bellwork: What dimension would a manufacturer need first?

Core Task: Add dimensions to a part drawing and check for missing information.

Evidence of Work / Notes:

Exit Ticket: What dimension did you add or revise?

Day 7 — Manufacturing Notes

Objective: I can write notes that clarify how a part should be made.

Bellwork: When are notes better than dimensions?

Core Task: Write notes for material, finish, edge quality, and safety constraints.

Evidence of Work / Notes:

Exit Ticket: What note would prevent confusion?

Day 8 — Bill of Materials

Objective: I can create a basic BOM.

Bellwork: Why do teams need a parts list?

Core Task: Create a BOM with part names, quantities, material, source, and notes.

Evidence of Work / Notes:

Exit Ticket: What part has the highest quantity?

Day 9 — Part Numbers and Revisions

Objective: I can track design changes using part numbers and revisions.

Bellwork: Why should design changes be recorded?

Core Task: Create part numbers and a revision table for a component.

Evidence of Work / Notes:

Exit Ticket: What changed in your revision?

Day 10 — Inspection Planning

Objective: I can plan how to inspect a manufactured part.

Bellwork: How do you know if a part was made correctly?

Core Task: Create an inspection checklist for critical dimensions and features.

Evidence of Work / Notes:

Exit Ticket: What is the most critical dimension?

Day 11 — Fabrication Planning

Objective: I can plan how a part will be manufactured.

Bellwork: Why should manufacturing be considered before building?

Core Task: Create a step-by-step fabrication plan.

Evidence of Work / Notes:

Exit Ticket: What step is most likely to cause an error?

Day 12 — Documentation Package Work Day 1

Objective: I can assemble manufacturing documentation.

Bellwork: What document would you give to a manufacturer first?

Core Task: Work on drawing, BOM, notes, and revision table.

Evidence of Work / Notes:

Exit Ticket: What is complete so far?

Day 13 — Documentation Package Work Day 2

Objective: I can revise documentation for clarity.

Bellwork: How can documentation fail even if the design is good?

Core Task: Peer review and revise the documentation package.

Evidence of Work / Notes:

Exit Ticket: What feedback did you use?

Day 14 — Final Documentation Check

Objective: I can check documentation for completeness.

Bellwork: What should you check before submission?

Core Task: Use the final checklist to inspect the package.

Evidence of Work / Notes:

Exit Ticket: What was missing or unclear?

Day 15 — Documentation Package Submission

Objective: I can submit and reflect on a manufacturing documentation package.

Bellwork: Why is documentation part of engineering?

Core Task: Submit documentation and complete reflection.

Evidence of Work / Notes:

Exit Ticket: What documentation skill improved most?

Unit 4 Submission Checklist

Item	Complete?
Measurement practice	☐
Tolerance notes	☐
Dimensioned drawing	☐
Manufacturing notes	☐
BOM	☐
Revision table	☐
Fabrication plan	☐
Documentation package	☐

Unit 4 Project Rubric

Category	Points
Technical accuracy and completeness	20
Engineering process and documentation	20
Aerospace connection and design intent	15
Quality of sketches, models, data, or drawings	20
Use of evidence, testing, or decision making	15
Reflection and professionalism	10
Total	100

Unit 4 Student Work Templates

Measurement Log

Purpose: Precision and inspection

Directions: Record measurements carefully with units. Compare actual measurements to target values and note possible sources of error.

Feature	Target Dimension	Measured Dimension	Difference / Notes
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Measurement reflection:

Manufacturing Drawing Template

Purpose: Orthographic / multiview sketching

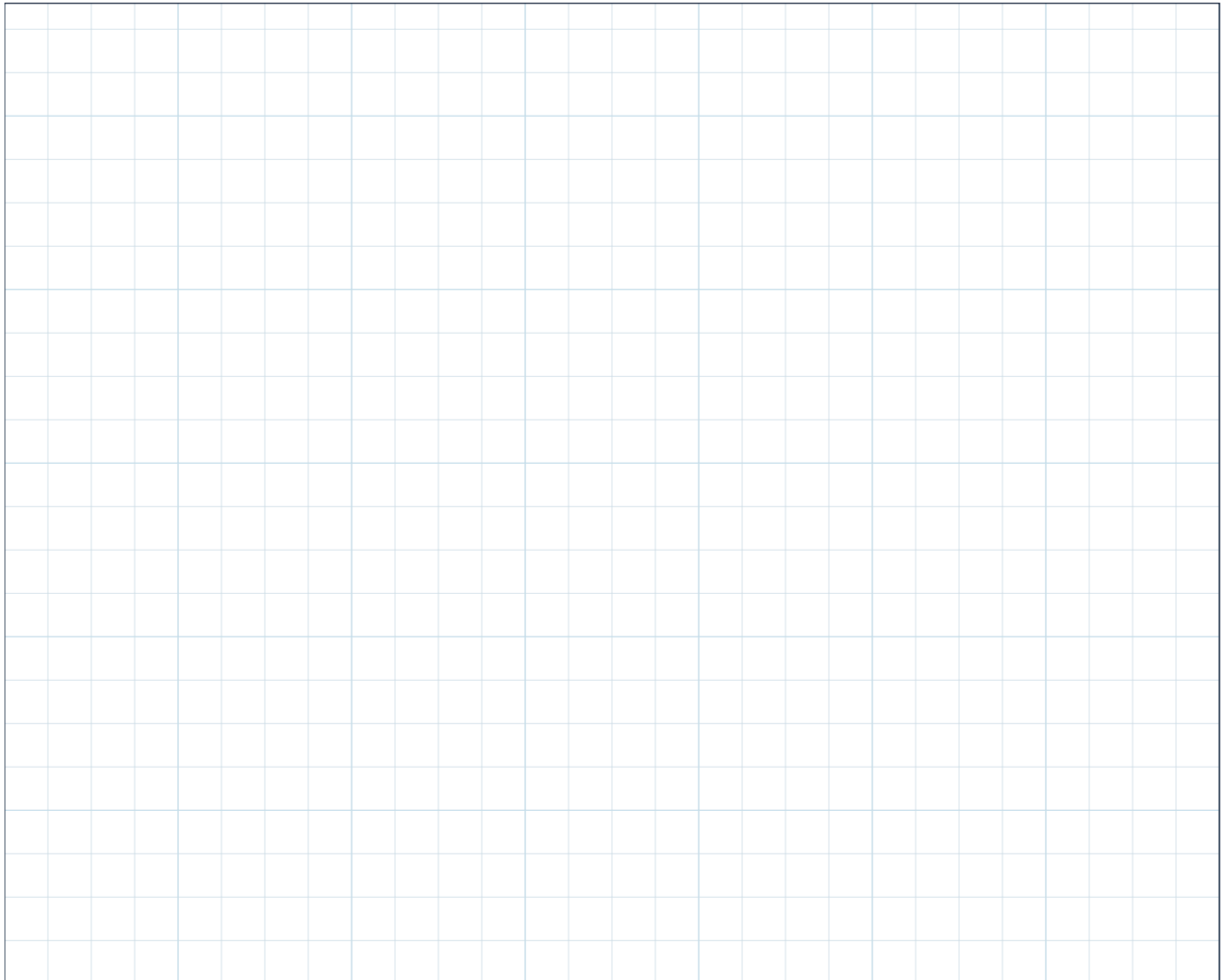
Directions: Use the orthographic grid to align the front, top, and right-side views. Show only the views requested. Use projection alignment and line conventions.

Part name: _____

Scale: _____

Main view: _____

Date: _____



Dimension / line convention notes: _____

Bill of Materials and Revision Template

Purpose: Documentation

Directions: List parts, materials, quantities, manufacturing methods, and revision notes so someone else could understand what changed.

Item	Part Name	Material / Process	Qty.	Notes
1				
2				
3				
4				
5				
6				

Rev.	Date	Change Made	Reason for Change
A			
B			
C			

Fabrication Planning Template

Purpose: Manufacturing preparation

Directions: Plan how your part or assembly would be made. Include materials, tool needs, safety concerns, and quality checks.

Material(s)**Tool/process**

Setup needs

Safety concern

Quality check

Fabrication sequence: